National University of Computer and Emerging Sciences



Sentence-Role Classification in Medical Abstracts: Fine-Tuned Biomedical Transformers vs. Classical Machine Learning on PubMed RCT

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Problem Statement:

Randomized controlled trial (RCT) abstracts are one of the main ways clinicians and researchers skim medical evidence, but many abstracts are unstructured, with no explicit separation of their parts. Labelling each sentence with its role (**background, objective, methods, results or conclusions**) makes abstracts faster to read and search, and is a building block for evidence-extraction and literature-review tools. Doing this by hand does not scale to the volume of published trials.

Fine-tuned biomedical transformer models such as Bio_ClinicalBERT have shown strong performance on biomedical text, while classical **bag-of-words pipelines** (TF-IDF with a linear or tree-based classifier) remain far cheaper to train and deploy. It is not yet clear whether the added complexity and compute cost of transformers is justified for classifying short sentences from medical abstracts.

Therefore, this project investigates whether fine-tuned biomedical transformers provide a **statistically significant improvement** over classical ML pipelines for automatically predicting the role of a sentence in a medical (RCT) abstract, and benchmarks TF-IDF + Logistic Regression / Random Forest / XGBoost against Bio_ClinicalBERT. The study is guided by the following research questions:

RQ1: How accurately can classical ML models (TF-IDF + classifier) classify the role of a sentence in a medical abstract (background, objective, methods, results, conclusions)?

RQ2: Do transformer-based models significantly outperform classical approaches on this task?

RQ3: What is the accuracy–efficiency trade-off (training time and inference cost) between the two paradigms?

Datasets:

Dataset	Role in Study	Key Feature	Source
PubMed 20k RCT	Primary dataset for training, validation and testing (predefined train / dev / test splits)	Real RCT abstract sentences labelled with their role (background, objective, methods, results, conclusions); 5-class text classification; about 180k / 30k / 30k sentences	Dernoncourt & Lee (2017); GitHub: Franck-Dernoncourt/pubmed-rct

Data Handling

- Files: `train.txt, dev.txt, test.txt` (PubMed_20k_RCT folder of the dataset repository). The raw files are immutable and excluded from version control (.gitignore); they are placed manually in `data/raw/`. Each line holds a label and a sentence, grouped by abstract.
- The dataset ships with predefined train / dev / test splits (about 180,040 / 30,212 / 30,135 sentences), which are used as provided; the splits are made by abstract, so no abstract appears in more than one split. Parsed copies are written to `data/processed/` (train.csv, val.csv, test.csv).
- Preliminary class distribution (training set): METHODS about 33%, RESULTS 32%, CONCLUSIONS 15%, BACKGROUND 12%, OBJECTIVE 8%. Full dataset statistics (sentence counts, class distribution, sentence-length statistics, missing values) will be reported after exploratory analysis in notebook 01. Macro F1 and class weighting will be used to handle the class imbalance.

Relevant Papers:

[1] PubMed 200k RCT: a Dataset for Sequential Sentence Classification in Medical Abstracts — Dernoncourt & Lee (2017). IJCNLP (Volume 2: Short Papers). Source of the dataset and its five sentence-role labels. https://arxiv.org/abs/1710.06071

[2] Publicly Available Clinical BERT Embeddings — Alsentzer et al. (2019) NAACL Clinical NLP Workshop. Source of Bio_ClinicalBERT, the biomedical transformer fine-tuned in this study. https://arxiv.org/abs/1904.03323

[3] BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding — Devlin et al. (2019) NAACL-HLT. Provides the transformer architecture baseline and the optional BERT comparison. https://arxiv.org/abs/1810.04805

[4] Pre-trained model: emilyalsentzer/Bio_ClinicalBERT The Hugging Face checkpoint that is fine-tuned in this study. https://huggingface.co/emilyalsentzer/Bio_ClinicalBERT

Project repository: https://github.com/Afras2077/esi-triage-ml-vs-transformers

Note: a project-specific paper does not yet exist; this study is itself the proposed contribution.

Project Scope

This project focuses on benchmarking **classical machine learning pipelines** against **fine-tuned transformer models** for classifying the role of each sentence in medical (RCT) abstracts. Specifically, it aims to investigate whether Bio_ClinicalBERT delivers a statistically significant improvement over TF-IDF based classifiers, and at what compute and inference cost.

Models to be Used

- **Logistic Regression, Random Forest, XGBoost**: Classical baselines using a TF-IDF vectoriser on cleaned sentence text.
- **Bio_ClinicalBERT**: Biomedical transformer fine-tuned via the HuggingFace Trainer on tokenised sentence text (max length 128; batch size 32; 5 epochs; learning rate 2e-5). BERT is an optional comparison.

Project Overview (Scope of Pipeline)

The project will cover the following high-level steps:

1. **Data Exploration**
    - Analyse class distribution, sentence length and abstract structure (notebook 01).
2. **Preprocessing**
    - Clean and tokenise text; load the predefined train / dev / test splits.
    - Inspect the TF-IDF vocabulary (notebook 02).
3. **Classical ML Baselines**
    - Train and evaluate Logistic Regression, Random Forest and XGBoost; save trained models (notebook 03).
4. **Transformer Fine-Tuning**
    - Fine-tune Bio_ClinicalBERT via the HuggingFace Trainer (notebook 04).
5. **Evaluation and Comparison**
    - Compare all models using Accuracy, Weighted F1, Macro F1 and Cohen's Kappa, with confusion matrices.
    - Run statistical significance tests and analyse the precision–efficiency trade-off (accuracy versus compute / inference cost) (notebook 05).

Out of Scope

- Sentence-context features (e.g., sentence position or neighbouring sentences); the baseline setting classifies each sentence from its text alone. Sequence-aware models are a possible extension.
- Deployment in real literature-search or clinical decision-support tools.
- Models beyond those listed (e.g., ClinicalLongformer is a possible future extension only).
- Clinical notes or patient records; the dataset consists of published abstracts, so results do not directly transfer to tasks such as ED triage.

Tech Stack and Project Organisation

- **Language / tools**: Python, Jupyter notebooks, scikit-learn, XGBoost, HuggingFace Transformers, NLTK, spaCy (en_core_web_sm).
- **Code structure**: modular `src/` package (data, features, models, evaluation, utils) with unit tests in `tests/`.
- **Configuration**: central `configs/config.yaml` for paths and hyperparameters.
- **Outputs**: `results/figures`, `results/metrics`, `results/reports`; serialised models in `models/`.